Chapter 10: Wave Optics

Formula Sheet + Tips & Tricks

Formula Sheet

I. Reflection and Refraction of Light Waves

Speed in a medium

$$v = \frac{c}{n}$$

c = speed in vacuum, n = refractive index of medium, v = speed in medium

Time to traverse a thickness

$$t = \frac{d}{v}$$

d = thickness of medium, v = speed of light in that medium

Wavelength in a medium

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$$

Frequency does not change on entering a new medium; only v and λ change

Relative refractive index

$$^{A}n_B = \frac{v_A}{v_B} = \frac{n_B}{n_A}$$

Refractive index of medium B with respect to medium A

Frequency–wavelength relation

$$f = \frac{c}{\lambda}$$

f = frequency, λ = wavelength in vacuum/air

II. Superposition of Waves

Intensity–amplitude relation

$$I \propto a^2$$

I = intensity, a = amplitude of the wave

Max. to min. intensity ratio

$$\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2$$

a_1, a_2 = amplitudes of the two superposing waves

Resultant intensity (phase form)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\cos\phi$$

ϕ = phase difference between the two waves at the point of interest

III. Young's Double Slit Experiment (YDSE)

Fringe width

$$\beta = \frac{\lambda D}{d}$$

D = slit-to-screen distance, d = slit separation, λ = wavelength

Position of nth bright fringe

$$x_n = \frac{n\lambda D}{d}$$

$n = 0, 1, 2, \ldots$ measured from the central bright fringe

Position of nth dark fringe

$$x_n = \frac{(2n-1)\lambda D}{2d}$$

$n = 1, 2, 3, \ldots$ measured from the central bright fringe

Path difference

$$\Delta x = d\sin\theta \approx \frac{xd}{D}$$

θ = angle subtended at the slits; valid for small θ

Phase difference

$$\phi = \frac{2\pi}{\lambda}\Delta x$$

Relates path difference Δx to phase difference ϕ

IV. Angular Fringe Width

Angular position of nth bright fringe

$$\theta_n = \frac{n\lambda}{d}$$

Small-angle form of x_n/D

Angular fringe width

$$\Delta\theta = \frac{\lambda}{d}$$

Angular separation between two consecutive bright (or dark) fringes

V. Diffraction at a Single Slit

Minima condition

$$a\sin\theta = n\lambda$$

$a =$ slit width, $n = 1, 2, 3, \ldots$

Secondary (subsidiary) maxima

$$a\sin\theta = (2n+1)\frac{\lambda}{2}$$

Approximate condition, $n = 1, 2, 3, \ldots$

Angular width of central maximum

$$2\theta = \frac{2\lambda}{a}$$

Angle between the first minima on either side of the central maximum

Linear width of central maximum

$$w = \frac{2\lambda D}{a}$$

$D =$ slit-to-screen distance (or focal length f of the focusing lens)

VI. Miscellaneous (Combined Relations)

Double-slit maxima inside single-slit envelope

$$N = \frac{2d}{e}$$

N = number of double-slit maxima within the central diffraction maximum, e = slit width, d = slit separation

Central maximum width using a lens

$$w = \frac{2f\lambda}{a}$$

Screen placed at focal plane of a convex lens of focal length f kept close to the slit

Tips, Tricks & Hacks

- Constructive interference: path difference $= n\lambda$; destructive: path difference $= (n + \frac{1}{2})\lambda$ — always compute path difference first, then just check which family it belongs to.
- Fringe width $\beta = \frac{\lambda D}{d}$: it increases with wavelength and distance to screen, decreases as slit separation increases — use this to instantly judge how fringe pattern changes if any one variable changes, without recomputing.
- If the whole YDSE apparatus is immersed in a medium of refractive index n, wavelength shrinks to λ/n, so fringe width shrinks to β/n — a very commonly tested modification.
- Diffraction from a single slit: central maximum is *twice* as wide as the secondary maxima, and its width $= \frac{2\lambda D}{a}$ — don't confuse this with the double-slit fringe width formula, they look similar but differ by that factor of 2 and by d vs a.
- Condition for minima in single-slit diffraction is $a \sin\theta = n\lambda$ (note: this is a minima condition here, unlike YDSE where $n\lambda$ gave maxima) — this sign/role flip is the single most common mix-up between the two topics.
- Resolving power of an optical instrument improves with *shorter* wavelength and *larger* aperture — this is why UV/electron microscopes resolve finer detail than visible-light ones.
- Polarisation only happens to transverse waves; sound (longitudinal) can never be polarised — a quick way to settle "can X be polarised" conceptual questions.
- Malus's law $I = I_0 \cos^2\theta$: intensity through a second polariser depends on $\cos^2$ of

the angle between the two polariser axes — remember it's $\cos^2$, not $\cos$, a very common careless error.